Absorbing the Unique Degenerate Fejer Shift into the Diagonal Ledger

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July 2026

Abstract

In the normalized TPC-130 Fejer inequality, the unique q=1,h=2 degenerate correlation is bounded by the same energy E as the diagonal. It changes the diagonal coefficient from 2 to at most 6 in units of 1/(H p), leaving only nondegenerate four-Mobius shifts. The classification is `ANALYTIC_REDUCTION_L1` and the verdict is `PROVED_DEGENERATE_SHIFT_ABSORPTION`. No program-positive L2 claim or endpoint exponent credit is made.

1 Target contract and source boundary

The target has six simultaneous axes: actual fixed-h_0 packet, source-locked named physical atom, every deterministic prefix, every deterministic scale, a fixed-X power at that atom, and actual active support. The value $h_0 = 2$ is source-backed data only. Repository hashes are integrity locks, not theorem evidence. TPC-193's declared seven-source corpus remains STOP-SCOPED [2].

2 Imported normalized inequality

For $1 \le H \le N$, $V = \sum |A_z| > 0$, $E = \sum |A_z|^2$, and $\mathfrak{p} = V^2/(NE)$, TPC-130 proves [1]

$$\frac{|S(\alpha)|^2}{V^2} \le 2\left\{\frac{1}{H\mathfrak{p}} + \frac{N}{HV^2}\left(\mathcal{O}_H(\alpha)\right)_+\right\},$$

where

$$\mathcal{O}_H(\alpha) = 2\Re\sum_{h=1}^{H-1}(1 - h/H)\mathrm{e}(-\alpha h)C(h).$$

If $V = 0$, then every $A_z = 0$ and the prefix is trivial; it is separated before this normalization.

Theorem 1 (Degenerate-shift absorption)**.** *Assume $3 \le H \le N$. For the unique degenerate cell $q = 1, h = 2$ identified in TPC-200 [3], Cauchy gives $|C(2)| \le E$. Its contribution to the normalized right-hand side is at most $4/(H\mathfrak{p})$. Together with the diagonal term, its total charge is at most*

$$\frac{6}{H\mathfrak{p}}.$$

Writing $\mathcal{O}_H = \mathcal{O}_H^{\mathrm{deg}} + \mathcal{O}_H^{\mathrm{nd}}$, the inequality $(x+y)_+ \le |x| + y_+$ gives the refined normalized bound

$$\frac{|S(\alpha)|^2}{V^2} \le \frac{6}{H\mathfrak{p}} + \frac{2N}{HV^2}\left(\mathcal{O}_H^{\mathrm{nd}}(\alpha)\right)_+.$$

Thus the degenerate term is absorbed into the diagonal feasibility condition by a fixed constant.

Proof. The $h = 2$ summand in $\mathcal{O}_H$ has absolute value at most $2(1 - 2/H)E \le 2E$. Multiplying by the outer factor $2N/(HV^2)$ gives at most $4NE/(HV^2) = 4/(H\mathfrak{p})$. The original diagonal term is $2/(H\mathfrak{p})$. □

3 Remaining gate

No cancellation is extracted from the degenerate term. After its absorption, the positive off-diagonal obligation is supported on the nonzero determinant rows of TPC-200. The missing theorem is therefore a uniform local or all-prefix estimate for that growing nondegenerate four-Möbius family.

4 Loss, level and scope ledger

The smallest missing literal theorem is

`UNIFORM_NONDEGENERATE_FOUR_MOBIUS_OFFDIAGONAL_POWER_BOUND`.

The next route is `NONDEGENERATE_FEJER_OFFDIAGONAL`. No new method cell is stopped in this paper.

The named-atom exponent credit is 0, while the endpoint requires a strict budget greater than 1/400; the ledger is unpaid. Block sums are not cumulative prefixes, symbolic packet atoms are not named production atoms, phase L^2 and almost-everywhere statements are not pointwise evaluation, and scoped method failures are not theorem or architecture failures.

5 Machine certificate

The adjacent canonical payload freezes the formula or finite witness, exact repository-source hashes, any explicitly non-hashed external locator record, six target axes, scoped stop, route state and claim firewall. Its recursive exact schema has closed objects, exact array positions and constant leaves. The checker recomputes the finite certificate and executes ten adversarial mutations.

References

[1] Liang Wang. Fejer four-sign h3 gate, 2026.

[2] Liang Wang. Literal fixed-atom candidate mechanism gate, 2026.

[3] Liang Wang. Four-form determinant resonance refinement, 2026.